Curriculum Vitae

AI & Software Engineer

Name: Loay Berzan

Phone: +49 15210351632

LinkedIn: linkedin.com/in/loay-berzan

Website: loayberzan.vercel.app

Address: Munich, Germany

E-Mail: loay.berzan@gmail.com

GitHub: github.com/loayasser

Work Eligibility: Valid German Student Visa

Education

Since 03/2025

Technische Hochschule Deggendorf (THD)

- M.Sc. Applied Computer Science.
- Focus: Business Application Re-Engineering, Machine Learning, AI & Data Science

10/2018 – 01/2023

Xiamen University, Malaysia Campus

- B.Eng. Electrical and Electronics Engineering.
- Focus: MCU Technology & Engineering, Data Communication Networks, Programming Language C

Practical Experience

10/2025 – 01/2026

Technische Hochschule Deggendorf, Deggendorf

First Semester Student Mentor

- Guide and support first-year M.Sc. Applied Computer Science students in academic integration, orientation, and navigating university systems.

10/2023 – 12/2024

Hijau Tech SDN BHD, Cyberjaya, Malaysia

Software Developer (Freelance)

- Built AI-driven dashboards and serverless backend workflows using React, Python, AWS Lambda, and Docker, delivering scalable, maintainable solutions with clean, modular codebases.
- Migrated legacy systems to a microservices architecture and established Git-based development workflows, improving team collaboration, deployment efficiency, and long-term code quality.

Skills

AI / ML

- TensorFlow, PyTorch, Keras, CNN, RAG Pipelines, Sentence-Transformers
- Pandas, Scikit-learn, Data Visualization, Computer Vision
- Time Series Forecasting, Random Forests

Programming

- Python
- JavaScript / HTML / CSS
- C++ / C / JSON / Jupyter Notebooks

Web & Backend

- React, FastAPI, REST APIs, AWS Lambda, API Gateway, Streamlit, n8n

DevOps & DBs

- Docker, Docker Compose, Git, CI/CD, Microservices
- PostgreSQL, pgvector, SQL, Linux

Low-Code / Other

- Mendix, MQTT, IoT (ESP32), Google Gemini API

Languages

- English: C1
- German: B1
- Arabic: Native

Computer Skills

Developer Tools

- VS Code, PyCharm, Jupyter Notebooks
- Git / GitHub / Docker
- Operating Systems (Windows, Linux)
- MS-Office (Word, Excel, etc..)

Projects

AI-Powered Document Research Assistant (RAG Pipeline) ***(On-Going)***

Technologies: Python, FastAPI, PostgreSQL, pgvector, Docker, PyTorch, Sentence-Transformers, Google Gemini API

- Architected an end-to-end RAG system using FastAPI to ingest, chunk, and semantically query unstructured PDF data, integrating Google Gemini API to synthesize accurate, hallucination-resistant answers.
- Engineered a local vectorization pipeline with PyTorch and Sentence-Transformers (all-MiniLM-L6-v2), storing 384-dimensional embeddings in PostgreSQL with pgvector for high-performance semantic similarity search.
- Containerized the full multi-service architecture using Docker Compose with custom volume management and health checks for a scalable, production-ready deployment.

German Traffic Sign AI Recognition

Technologies: Python, TensorFlow, Keras, Streamlit, CNN, GTSRB Dataset

- Engineered a custom CNN classifier achieving 92% validation accuracy across 43 traffic sign categories, designed to maintain reliable performance under varying lighting and image quality conditions.
- Performed dataset preparation and preprocessing on the GTSRB dataset, including image normalization, augmentation, and class balancing to improve model generalization.
- Live Demo: traffic-sign-recognizergit-hzyappvcpygdvkykdmtgeyp.streamlit.app| GitHub: github.com/loayasser/Traffic-Sign-Recognizer

AI-Assisted Manufacturing Insights Dashboard

Technologies: Mendix, REST APIs, AI Services

- Built a low-code application that reliably processes and visualizes manufacturing KPI data in real time, enabling production teams to identify bottlenecks and inefficiencies at a glance.
- Integrated AI services via REST APIs to automatically generate actionable performance summaries, reducing manual analysis time and supporting faster, data-driven decision-making.

IoT Weather Station (ESP32)

Technologies: ESP32, DHT11, BMP180, MQTT, ThingSpeak, LCD

- Developed a reliable real-time environmental monitoring system using ESP32, with continuous sensor data acquisition, cloud sync via MQTT, and local LCD fallback display ensuring zero data-loss during connectivity interruptions.

Additional Qualifications

Achievements

- 24-Hour Hackathon (United AI Club x MalMachen): Built an AI-powered email distribution system to automate corporate communication workflows under time pressure.
- Renewable Energy Hackathon: Engineered a prototype hydroelectric power generation system integrating IoT and AI for real-time monitoring and efficiency optimization.

Munich, 25.3.2026